MAHAM ZAFAR

Software Engineer / AI Engineer | Expert in LLM & RAG Architectures | Agentic AI / Generative AI | Full-Stack Development

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PROFESSIONAL SUMMARY

Results-driven Software Engineer and AI Engineer with enterprise-level experience in Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and R2R Agentic pipelines, complemented by hands-on full-stack web development experience with Vue.js, Django, and SQL. Proven ability to architect and deploy production-ready Agentic RAG systems, fine-tune transformer-based models, build intelligent automation workflows, and deliver complete web applications. Passionate about building scalable, data-driven AI solutions that solve real-world problems.

TECHNICAL SKILLS

AI / Machine Learning: LLM Fine-Tuning & Evaluation, NLP, Feature Engineering, Generative AI, RAG, Agentic Pipelines
Programming & Frameworks: Python (primary), C++, Java, JavaScript, FastAPI, Django, Vue.js, Node.js / Express, React, Pandas, NumPy, JSON
Data & Databases: Data Cleaning / Preprocessing, SQL, MySQL, MongoDB, Oracle SQL
Tools & Platforms: Git, GitHub, VS Code, Google Colab
Methodologies: Agile / Scrum, SDLC, Prompt Engineering, RAG Pipeline Development, Agentic AI Workflows, REST API Integration, AI Automation

PROFESSIONAL EXPERIENCE

Software Engineer Intern — Rasant Solutions *Sep 2026 – Oct 2026*

- Completed a two-month internship as a full-stack developer, building and maintaining web application features using Vue.js on the frontend and Django on the backend.
- Developed responsive, component-based user interfaces in Vue.js and connected them to Django REST endpoints for dynamic data handling.
- Designed and wrote SQL queries, managed database models, and worked with relational data to support application features and reporting.
- Collaborated with the development team on debugging, code reviews, and feature delivery within an Agile workflow, using Git for version control.

Software Engineer Intern — NASTP (FYP) *Sep 2025 – July 2026*

Rawalpindi, Pakistan

- Developed AI-powered document processing solutions using Python, FastAPI, and Hugging Face Transformers, focusing on automated extraction of structured information from scanned and handwritten documents.
- Fine-tuned and evaluated Donut Vision-Language models on a domain-specific dataset, improving structured field extraction from NASTP leave-application forms.
- Built and integrated RESTful APIs using FastAPI and Node.js/Express to connect AI inference services with web-based applications and data storage.
- Contributed to the development of an end-to-end Generative AI and intelligent document processing workflow, including document ingestion, processing, structured JSON extraction, and result management.
- Collaborated within an Agile/Scrum environment, translating project requirements into development tasks and integrating AI components with the overall system architecture.

Data Analyst Intern — Punjab Group Of Colleges *Jun 2025 – Aug 2025*

KabirWala, Pakistan

- Analyzed student attendance, academic performance, and enrollment data to identify patterns, trends, inconsistencies, and meaningful insights.
- Used Python, Pandas, NumPy, and Matplotlib for data cleaning, preprocessing, analysis, and visualization.
- Prepared analytical reports and visualizations to communicate student-related insights and support data-driven decision-making.
- Collaborated with team members on student data analysis and reporting tasks, strengthening practical skills in Python, data analytics, and visualization.

Exploratory Data Analyst Intern — AICP *Jan 2025 – Mar 2025*

Remote, Lahore, Pakistan

- Performed exploratory data analysis (EDA) on datasets to identify patterns, trends, inconsistencies, and meaningful insights.
- Used Python, Pandas, NumPy, and Matplotlib for data cleaning, preprocessing, analysis, and visualization.
- Prepared analytical reports and visualizations to communicate data-driven insights and support decision-making.
- Collaborated with team members on data analysis tasks, contributing to data-driven projects and technical research while strengthening practical skills in Python and analytics.

PROJECTS

DonutDoc – Intelligent Document Processing (IDP) System (GitHub link) *Final Year Project Sponsored by NASTP*

Python · Donut · Hugging Face · FastAPI · Node.js · Express.js · React · MongoDB · Google Colab · CUDA

- A full-stack AI-powered document processing system that uses the OCR-free Donut Vision-Language Model to automatically understand and extract structured information from scanned and handwritten NASTP leave-application forms with 95% accuracy.
- Fine-tuned the Donut model (Swin Transformer encoder + BART decoder) on a domain-specific dataset of handwritten and synthetic forms to perform structured field extraction without traditional OCR.
- Developed a FastAPI-based AI inference service and integrated it with a Node.js/Express backend and React frontend, enabling document upload, processing, structured JSON generation, and result management.
- Implemented support for JPG, JPEG, PNG, and multi-page PDF documents, along with document history, processing status, authentication, and archived results for an end-to-end intelligent document workflow.

AI Interview Coach (GitHub link)

Python · Streamlit · Google Gemini API · SQLite · Plotly · FPDF

- Developed an AI-powered interview preparation platform that uses Google Gemini to conduct interactive HR, technical, and behavioral interviews and evaluate candidate responses.
- Implemented AI-driven response analysis and scoring to generate personalized feedback, performance insights, and improvement suggestions.
- Built user authentication and interview history using SQLite, with Plotly-based performance visualization and automated PDF report generation.

DocuMind AI (GitHub link)

Python · Streamlit · Google Gemini · RAG · ChromaDB · LangChain

- Engineered a multi-document RAG system that processes PDF, DOCX, PPTX, and TXT files and enables context-aware question answering over uploaded documents.
- Implemented document chunking, vector embeddings, semantic retrieval, and contextual generation to provide relevant responses grounded in the user's documents.
- Added intelligent document capabilities including summarization, document comparison, quizzes, flashcards, and PDF report generation.

Facial Expression Recognition (GitHub link)

Python · TensorFlow · Keras · CNN · OpenCV

- Developed a deep learning-based facial expression recognition system using Convolutional Neural Networks (CNNs) to classify emotions from facial images.
- Implemented image preprocessing, model training, validation, and performance evaluation using TensorFlow and Keras.
- Integrated OpenCV for image processing and real-time facial expression prediction.

EDUCATION

Bachelor of Software Engineering — University of Engineering and Technology Taxila *2022 – 2026*

- CGPA: 3.68 / 4.0

HONORS AND AWARDS

- PEC Merit Scholarship — Awarded for outstanding academic performance in Matriculation.
- Full Merit-Based Scholarship — Punjab Group of Colleges.
- 3rd Position, Multan Board — Secured 3rd position in Intermediate examinations.
- Prime Minister’s Merit-Based Laptop Award — Awarded a laptop in recognition of academic excellence.

CERTIFICATIONS

- Crash Course on Python Programming — Google / Coursera (2023)
- Supervised Machine Learning & Advanced Learning Algorithms — Stanford University / Coursera, Andrew Ng (2024)
- Operating System Interaction with Python — Google / Coursera (2024)
- ChatGPT Prompt Engineering & LangChain for LLMs — DeepLearning.AI (2025)